9.1 Right Where We Left

Let’s recap where we left off in previous chapters: we are designing software for an e-commerce company, an application to find the closest warehouse selling a given product, for any point on a very large map: see figure 9.1 to visualize. To have a ballpark idea of the kind of scale we need, we want to serve millions of clients per day across the Country, taking products from thousands of warehouses, also spread across the whole map.

In section 8.2, we have already established that a brute-force solution where we skim through the whole list of points to compare differences can’t work for a live solution.

We have also seen how the multidimensional structure of the data prevents us from using the basic solutions we have seen in the first part of the book, from heaps to hash maps.